IA Mathematics for Natural Sciences B:
Examples Sheet 2 - Triple products, vector area, and polar coordinates

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Questions marked with a $(*)$ are difficult and should not be attempted at the expense of the other questions.

Vector products: perpendicular projections

1. Draw a diagram showing that the magnitude of the component of $\mathbf{v}$ perpendicular to $\mathbf{u} \neq \mathbf{0}$ is $|\mathbf{v} \times \hat{\mathbf{u}}|$. Show further that the projection of $\mathbf{v}$ perpendicular to $\mathbf{u}$ is $\hat{\mathbf{u}} \times (\mathbf{v} \times \hat{\mathbf{u}})$. Using these results:
   (a) Show that an equation of a line passing through the point $\mathbf{a}$, in the direction $\mathbf{v} \neq \mathbf{0}$, is $\mathbf{r} \times \mathbf{v} = \mathbf{m}$, where $|\mathbf{m}|/|\mathbf{v}|$ is the shortest distance to the origin. Find the equations of the lines L_1, L_2 from Sheet 1 Question 6 in the form $\mathbf{r} \times \mathbf{v} = \mathbf{m}$; hence, find their shortest distances to the origin.
   (b) Explain why an equation of the plane through $\mathbf{a}$ with normal $\mathbf{n} \neq \mathbf{0}$ is $\hat{\mathbf{n}} \times ((\mathbf{r} - \mathbf{a}) \times \hat{\mathbf{n}}) = \mathbf{r} - \mathbf{a}$.
2. By considering appropriate perpendicular projections, show that:
   (a) the shortest distance from a point $\mathbf{p}$ to a line passing through a point $\mathbf{a}$ with direction vector $\mathbf{v}$ is given by $|(\mathbf{p} - \mathbf{a}) \times \hat{\mathbf{v}}|$, and the closest point on the line to the point is $\mathbf{p} - \hat{\mathbf{v}} \times ((\mathbf{p} - \mathbf{a}) \times \hat{\mathbf{v}})$.
   (b) the shortest distance from a point $\mathbf{p}$ to a plane passing through a point $\mathbf{a}$ with normal $\mathbf{n}$ is given by $\sqrt{|\mathbf{p} - \mathbf{a}|^2 - |(\mathbf{p} - \mathbf{a}) \times \hat{\mathbf{n}}|^2}$.

   Hence, use the vector product to find the shortest distance from the point $P = (1, 0, 1)$ to the planes P_1, P_2 and the lines L_1, L_2 from Sheet 1 Questions 6 and 7.

Vector and scalar triple products

3. State *Lagrange's formula* for the *vector triple product* $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$; now, learn it off by heart. Give an example of three vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ such that $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) \neq (\mathbf{a} \times \mathbf{b}) \times \mathbf{c}$. Using Lagrange's formula:
   (a) prove *Jacobi's identity*, $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) + \mathbf{b} \times (\mathbf{c} \times \mathbf{a}) + \mathbf{c} \times (\mathbf{a} \times \mathbf{b}) = \mathbf{0}$;
   (b) verify that the equation of the plane in Question 1(b) is equivalent to $(\mathbf{r} - \mathbf{a}) \cdot \mathbf{n} = 0$.
4. Consider the equation $\mathbf{a} \times \mathbf{r} + \lambda \mathbf{r} = \mathbf{c}$, for the variable $\mathbf{r}$, where $\lambda \neq 0$ and $\mathbf{a}, \mathbf{c} \in \mathbb{R}^3$ are constant.
   (a) By taking the scalar product of the equation with $\mathbf{a}$, show that $\lambda(\mathbf{r} \cdot \mathbf{a}) = \mathbf{c} \cdot \mathbf{a}$.
   (b) By taking the vector product of the equation with $\mathbf{a}$, and using (a), find the equation's solution.
   (c) Does the equation have any solutions when $\lambda = 0$ and $\mathbf{a} \cdot \mathbf{c} \neq 0$? What if $\lambda = 0$, and $\mathbf{a} \cdot \mathbf{c} = 0$? Find the solutions, if they exist, in each case. [*Hint: what does (a) tell you about $\boldsymbol{r} \cdot \boldsymbol{a}$ in these cases?*]
5. (a) Solve $2\mathbf{r} + \hat{\mathbf{n}} \times \mathbf{r} + \hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \mathbf{r})^2 = \mathbf{a}$ for $\mathbf{r}$, where $\hat{\mathbf{n}}$ is a unit vector, and $\hat{\mathbf{n}} \cdot \mathbf{a} = -1$;
   (b) Solve $\mathbf{r} = \mathbf{a} - (\mathbf{b} \cdot \mathbf{r})\mathbf{b} - \mathbf{b} \times \mathbf{r}$ for $\mathbf{r}$, where $\mathbf{a}, \mathbf{b}$ are constant vectors.
6. (a) Define the *scalar triple product* $[\mathbf{a}, \mathbf{b}, \mathbf{c}]$ of three vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$. With a diagram, show that the scalar triple product $[\mathbf{a}, \mathbf{b}, \mathbf{c}]$ represents the *signed* volume of the parallelepiped formed by $\mathbf{a}, \mathbf{b}, \mathbf{c}$.
   (b) Hence, explain why: (i) the scalar triple product is antisymmetric on odd permutations, and symmetric on even permutations; (ii) three vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ form a basis if and only if $[\mathbf{a}, \mathbf{b}, \mathbf{c}] \neq 0$.
   (c) Let $\mathbf{a} = (0, \frac{1}{2}, \frac{1}{2})$, $\mathbf{b} = (\frac{1}{2}, 0, \frac{1}{2})$ and $\mathbf{c} = (\frac{1}{2}, \frac{1}{2}, 0)$. Compute $[\mathbf{a}, \mathbf{b}, \mathbf{c}]$, $[\mathbf{b}, \mathbf{a}, \mathbf{c}]$ and $[\mathbf{b}, \mathbf{c}, \mathbf{a}]$, and comment on how they are related in light of (b)(i). Do the vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ form a basis?
7. Simplify the scalar triple products $(\mathbf{a} + \mathbf{b}) \cdot (\mathbf{b} + \mathbf{c}) \times (\mathbf{c} + \mathbf{a})$ and $(\mathbf{a} \times \mathbf{b}) \cdot [(\mathbf{b} \times \mathbf{c}) \times (\mathbf{c} \times \mathbf{a})]$.

8. (a) Given the vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are orthonormal, derive expressions for the coefficients α, β, γ in the formula $\mathbf{d} = \alpha\mathbf{a} + \beta\mathbf{b} + \gamma\mathbf{c}$. Hence express $(2, 3, 4)$ in terms of the basis $\{(1, 1, 0), (1, -1, 0), (0, 0, 1)\}$.
   (b) If instead the vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are *not* orthonormal, derive expressions for the coefficients α, β, γ in the formula $\mathbf{d} = \alpha\mathbf{a} + \beta\mathbf{b} + \gamma\mathbf{c}$. Hence express $(1, 1, 1)$ in terms of the basis $\{(1, 2, 1), (0, 0, 1), (2, -1, 1)\}$.
   (c) Assuming $\mathbf{a}, \mathbf{b}, \mathbf{c}$ form a basis, we define the *reciprocal basis* to $\mathbf{a}, \mathbf{b}, \mathbf{c}$ to be the set of vectors:

$$\mathbf{A} = (\mathbf{b} \times \mathbf{c})/[\mathbf{a}, \mathbf{b}, \mathbf{c}], \qquad \mathbf{B} = (\mathbf{c} \times \mathbf{a})/[\mathbf{a}, \mathbf{b}, \mathbf{c}], \qquad \mathbf{C} = (\mathbf{a} \times \mathbf{b})/[\mathbf{a}, \mathbf{b}, \mathbf{c}].$$

   Show that $\mathbf{A} \cdot \mathbf{a} = \mathbf{B} \cdot \mathbf{b} = \mathbf{C} \cdot \mathbf{c} = 1$, and $\mathbf{A} \cdot \mathbf{b} = \mathbf{A} \cdot \mathbf{c} = \mathbf{B} \cdot \mathbf{a} = \mathbf{B} \cdot \mathbf{c} = \mathbf{C} \cdot \mathbf{a} = \mathbf{C} \cdot \mathbf{b} = 0$. Find the reciprocal basis to the basis in part (b), and explain how to use it to find the coefficients α, β, γ.

Vector area

9. (a) Define the *vector area* $\mathbf{A}$ of a connected, flat surface that has area A and unit normal $\hat{\mathbf{n}}$. What is the convention that is normally used when choosing the direction of the normal?
   (b) Determine the vector areas of the: (i) parallelogram spanned by $\mathbf{a}, \mathbf{b}$; (ii) triangle spanned by $\mathbf{a}, \mathbf{b}$.
   (c) The velocity $\mathbf{u} = (1, 0, 1)^T$ of a fluid is uniform everywhere. Compute the volume passing the triangle spanned by $\mathbf{a} = (0, 0, 1)^T, \mathbf{b} = (0, 1, 0)^T$ per unit time. [*This is called the* velocity flux.]

10. (a) Define the vector area $\mathbf{A}$ of a connected, non-flat surface composed piecewise of flat surfaces of areas $A_1, ..., A_k$ with unit normals $\hat{\mathbf{n}}_1, ..., \hat{\mathbf{n}}_k$ respectively. What do we require about the normals?
    (b) Find the vector area $\mathbf{S}$ of the surface S made up of the two triangles OAB, BCO, where $A = (1, 0, 0)$, $B = (1, 1, 1)$, $C = (0, 2, 0)$. Is $|\mathbf{S}|$ equal to the surface area of S?

11. Give a very general outline of how the concept of vector area might be extended to *curved* surfaces S; hence, explain why we expect the vector area of a *closed* surface to be $\mathbf{0}$. Thus, find the vector areas of:
    (a) a square-based pyramid with vertices $(0, 0, 0)$, $(2, 0, 0)$, $(2, 2, 0)$, $(0, 2, 0)$, $(1, 1, 1)$, with base removed;
    (b) the unit hemisphere $x^2 + y^2 + z^2 = 1, z \geq 0$.

12. (a) Let $\mathbf{S}$ be the vector area of the surface S. Prove that the area of the projection (or 'shadow') of the surface S onto the plane with unit normal $\hat{\mathbf{m}}$ is $|\mathbf{S} \cdot \hat{\mathbf{m}}|$.
    (b) Using the result from part (a), compute the area *and* the vector area of the projection of the square with vertices $(0, 0, 0)$, $(2, 0, 0)$, $(2, 2, 0)$ and $(0, 2, 0)$ onto a plane with normal $\mathbf{m} = (0, -1, 1)$.

Polar coordinate systems

13. Draw convincing diagrams defining plane polar, cylindrical, and spherical coordinates. Hence, find the cylindrical polar and spherical polar coordinates of the point $(3, 4, 5)$.

14. (a) In 2D Cartesians, a circle is specified by $(x - 1)^2 + y^2 = 1$. Find its equation in polars.
    (b) In 3D Cartesians, a sphere is specified by $(x - 1)^2 + y^2 + z^2 = 1$. Find its equation in sphericals.

15. Let $a > 0$ be a constant. Describe the following loci:
    (a) (i) $\phi = a$; (ii) $r = \phi$, in plane polar coordinates;
    (b) (i) $z = a$; (ii) $r = a$; (iii) $r = a$ and $z = \phi$, in cylindrical polar coordinates;
    (c) (i) $\theta = a$; (ii) $\phi = a$; (iii) $r = a$; (iv) $r = \theta = a$, in spherical polar coordinates.

16. Consider a point with position vector $\hat{\mathbf{n}}$ on the unit sphere S centred on the origin.
    (a) Show that $\hat{\mathbf{n}} = (\sin(\theta)\cos(\phi), \sin(\theta)\sin(\phi), \cos(\theta))^T$, where (r, θ, ϕ) are polar coordinates.
    (b) Show that the vector area $d\mathbf{S}$ of a small patch near $\hat{\mathbf{n}}$, subtending a small angle $d\theta$ in the θ direction, and a small angle $d\phi$ in the ϕ direction, is given *approximately* by $d\mathbf{S} = \hat{\mathbf{n}} \sin(\theta) d\theta d\phi$.
    (c) $(*)$ Hence, use integration to verify that the vector area of the unit sphere is $\mathbf{0}$.